G0211	Quiz 2	Marks: /40

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Instruction: Answer FIVE questions in the provided space. Justify your answer in each question, and in order to get a full mark, you must demonstrate all necessary steps.

Question 1 (10 marks)

Evaluate the definite integral $\int_5^6 \frac{x+11}{x^2-x-6} dx$. Round your final answer to four decimal places.

Solution:

$x^2-x-6=(x-3)(x+2)$

We can have $\frac{x+11}{x^2-x-6}=\frac{A}{x-3}+\frac{B}{x+2}=\frac{A(x+2)+B(x-3)}{x^2-x-6}$

$x+11=(A+B)x+2A-3B$

$\Rightarrow \begin{cases} A+B=1 \\ 2A-3B=11 \end{cases} \Rightarrow \begin{cases} A=\frac{14}{5} \\ B=-\frac{9}{5} \end{cases}$

$$\begin{aligned}
\int_5^6 \frac{x+11}{x^2-x-6}dx &= \int_5^6 \left(\frac{14}{5(x-3)}-\frac{9}{5(x+2)}\right)dx \\
&= \frac{14}{5}\int_5^6 \frac{1}{x-3}dx - \frac{9}{5}\int_5^6 \frac{1}{x+2}dx \\
&= \frac{14}{5}\left[\ln|x-3|\right]_5^6 - \frac{9}{5}\left[\ln|x+2|\right]_5^6 \\
&= \frac{14}{5}\left[\ln(6-3)-\ln(5-3)\right]-\frac{9}{5}\left[\ln(6+2)-\ln(5+2)\right] \\
&= \frac{14}{5}\ln\frac{3}{2}-\frac{9}{5}\ln\frac{8}{7} \\
&\approx 0.8951
\end{aligned}$$

$\Rightarrow \int_5^6 \frac{x+11}{x^2-x+6}dx = 0.8951$

Question 2 (5 marks)

Use a trigonometric substitution to evaluate the indefinite integral $\int \frac{\sqrt{9-x^2}}{x^2}\,dx$.

Solution:

$$x = 3\sin\theta \ (a=3),\ dx = 3\cos\theta\, d\theta$$

$$\sqrt{9-x^2} = \sqrt{9-9\sin\theta} = 3\cos\theta$$

$$\int \sqrt{9-x^2}\,dx = \int \frac{3\cos\theta}{(3\sin\theta)^2} \cdot 3\cos\theta\, d\theta$$

$$= \int \frac{9\cos^2\theta}{9\sin^2\theta}\, d\theta$$

$$= \int \frac{\cos^2\theta}{\sin^2\theta}\, d\theta$$

$$= \int \cot^2\theta\, d\theta$$

$$= \int (\csc^2\theta - 1)\, d\theta$$

$$= -\cot\theta - \theta + C$$

$$x = 3\sin\theta \Rightarrow \sin\theta = \frac{x}{3} \Rightarrow \theta = \arcsin\frac{x}{3}$$

$$\cot\theta = \frac{\sqrt{9-x^2}}{x}$$

$$\Rightarrow \int \frac{\sqrt{9-x^2}}{x^2}\,dx = -\frac{\sqrt{9-x^2}}{x} - \arcsin\frac{x}{3} + C$$

Question 3 (5 marks)

Use an appropriate substitution to solve the indefinite integral $\int xe^{2x^2}\,dx$. [5]

Solution:

let $u = 2x^2$

$\frac{du}{dx} = 4x \Rightarrow du = 4x\,dx \Rightarrow x\,dx = \frac{1}{4}du$

$\int xe^{2x^2}dx = \int e^u \cdot \frac{1}{4}du = \frac{1}{4}\int e^u du$

$\frac{1}{4}\int e^u du = \frac{1}{4}e^u + C$

substitute back $u = 2x^2$

$\Rightarrow \int xe^{2x^2}dx = \frac{1}{4}e^{2x^2} + C$

Question 4 (12 marks)

Region D is a region in the xy-plane which is bounded by the parabola $4x = y^2$ and the straight line $y = 2x - 4$.

a) Find the area of the region D. [4]

b) Compute the volume of the solid generated by revolving the region D about the horizontal line $y = -2$ through Cylindrical Shell method. [8]

Solution:

(a) $4x = y^2 \Rightarrow x = \frac{y^2}{4}$

$y = 2x - 4 \Rightarrow x = \frac{y+4}{2}$

$\frac{y^2}{4} = \frac{y+4}{2} \Rightarrow 2y^2 = 4y + 16$

$\Rightarrow y^2 - 2y - 8 = 0$

$\Rightarrow (y-4)(y+2) = 0$

① $y = 4$, so $x = 4$

② $y = -2$, so $x = 1$

So the region D is bounded between $y=-3$ and $y=4$.

$$\text{Area} = \int_{-2}^{4} \left(\frac{y}{2}+2\right) - \left(\frac{y^2}{4}\right) dy$$

$$= \left[\frac{y^2}{4} + 2y - \frac{y^3}{12}\right]_{-2}^{4}$$

$$= \left(\frac{4^2}{4} + 2\cdot 4 - \frac{4^3}{12}\right) - \left[\frac{(-2)^2}{4} + 2\cdot(-2) - \frac{(-2)^3}{12}\right]$$

$$= \frac{20}{3} - \left(-\frac{7}{3}\right)$$

$$= 9$$

$\Rightarrow$ the area of region D is 9.

(b) $r = y - (-2) = y + 2$

$h = \left(\frac{y}{2}+2\right) - \left(\frac{y^2}{4}\right) = -\frac{y^2}{4} + \frac{y}{2} + 2$

$$V = 2\pi \int_{-2}^{4} (y+2)\left(-\frac{y^2}{4} + \frac{y}{2} + 2\right) dy$$

$$= 2\pi \int_{-2}^{4} \left(-\frac{y^3}{4} + \frac{y^2}{2} + 2y - \frac{2y^2}{4} + \frac{2y}{2} + 4\right) dy$$

$$= 2\pi \int_{-2}^{4} \left(-\frac{y^3}{4} + 3y + 4\right) dy$$

$$= 2\pi \left[-\frac{y^4}{16} + \frac{3y^2}{2} + 4y\right]_{-2}^{4}$$

$$= 2\pi \left[\left(-\frac{4^4}{16} + \frac{3\cdot 4^2}{2} + 4\cdot 4\right) - \left(-\frac{(-2)^4}{16} + \frac{3\cdot(-2)^2}{2} + 4\cdot(-2)\right)\right]$$

$$= 2\pi\left[24 - (-3)\right]$$

$$= 54\pi$$

Question 5 (8 marks)

A region is enclosed by the curves $y = x$ and $y = x^2$. The region is rotated about the line $y = 2$. Find the volume of the resulting solid through Washer method.

Solution:

$x = x^2 \Rightarrow x^2 - x = 0 \Rightarrow x(x-1) = 0$

① $x = 0$, so $y = 0$

② $x = 1$, so $y = 1$

So the region is bounded between $y=0$ and $y=1$.

we have 2 rotation radus about $y=2$

$r_1 = 2 - x^2$

$r_2 = 2 - x$

$V = \pi \int_0^1 [(2-x^2)^2 - (2-x)^2] dx$

$= \pi \int_0^1 [(4 - 4x^2 + 4x^2) - (4 - 4x + x^2)]$

$= \pi \int_0^1 (x^4 - 5x^2 + 4x)$

$= \pi \left[\frac{x^5}{5} - \frac{5x^3}{3} + 2x^2 \right]_0^1$

$= \pi \left[\left(\frac{1^5}{5} - \frac{5 \cdot 1^3}{3} + 2 \cdot 1^2 \right) - \left(\frac{0^5}{5} - \frac{5 \cdot 0^3}{3} + 2 \cdot 0^2 \right) \right]$

$= \pi \left(\frac{8}{15} - 0 \right)$

$= \frac{8}{15} \pi$

$\Rightarrow$ the volume of the solid is $\frac{8}{15}\pi$,

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